

Seminar Nr. 3, Probability and Statistics

Probabilistic Models

Theory Review

Binomial Model: The probability of k successes in n Bernoulli trials, with probability of success p ($q = 1 - p$), is

$$P(n, k) = C_n^k p^k q^{n-k}, \quad k = \overline{0, n}.$$

Hypergeometric Model: The probability that in n trials, we get k successes out of n_1 and $n - k$ failures out of $N - n_1$ ($0 \leq k \leq n_1, 0 \leq n - k \leq N - n_1$), is

$$P(n; k) = \frac{C_{n_1}^k C_{N-n_1}^{n-k}}{C_N^n}.$$

Poisson Model: The probability of k successes ($0 \leq k \leq n$) in n trials, with probability of success p_i in the i^{th} trial ($q_i = 1 - p_i$), $i = \overline{1, n}$, is

$$P(n; k) = \sum_{1 \leq i_1 < \dots < i_k \leq n} p_{i_1} \dots p_{i_k} q_{i_{k+1}} \dots q_{i_n}, \quad i_{k+1}, \dots, i_n \in \{1, \dots, n\} \setminus \{i_1, \dots, i_k\}$$

= the coefficient of x^k in the polynomial expansion $(p_1x + q_1)(p_2x + q_2) \dots (p_nx + q_n)$.

Pascal (Negative Binomial) Model: The probability of the n^{th} success occurring after k failures in a sequence of Bernoulli trials with probability of success p ($q = 1 - p$), is

$$P(n; k) = C_{n+k-1}^{n-1} p^n q^k = C_{n+k-1}^k p^n q^k.$$

Geometric Model: The probability of the 1^{st} success occurring after k failures in a sequence of Bernoulli trials with probability of success p ($q = 1 - p$), is

$$p_k = pq^k.$$

1. Five percent of computer parts produced by a certain supplier are defective. What is the probability that a sample of 16 parts contains

- a) exactly 3 defective parts (ev. A)?
- b) more than 3 defective parts? (ev. B)?
- c) at least one defective part (ev. C)?
- d) less than 3 defective parts (ev. D)?

2. There are 200 seats in a theater, 10 of which are reserved for the press. 150 people come to the show one night, and are seated randomly. What is the probability of all the seats reserved for the press to be occupied (ev. A)?

3. Among 10 laptop computers, seven are good, the rest have defects. Unaware of this, a customer buys 5 laptops.

- a) What is the probability of exactly 2 defective ones among them (ev. A)?
- b) Knowing that *at least* 2 purchased laptops are defective, what is the probability that *exactly* 2 are defective (ev. B)?

4. A computer program is tested by 5 independent tests. If there is an error, these tests will detect it with probabilities 0.1, 0.2, 0.3, 0.4 and 0.5, respectively. Suppose that the program contains an error. What is the probability that it will be found by

- a) at least one test (ev. A)?
- b) more than two tests (ev. B)?
- c) all five tests (ev. C)?

5. In a public library, 1 out of 10 people using the computers do not close Windows properly. What is the probability that Windows is closed properly only by the 3rd user (event A)?

6. An engineer tests the quality of produced computers. Suppose that 5% of computers have defects and defects occur independently of each other. Find the probability

- a) of exactly 3 defective computers in a shipment of 20 (ev. A);
- b) that the engineer has to test at least 5 computers in order to find 2 defective ones (ev. B).

7. (Banach's Problem). A person buys 2 boxes of aspirin, each containing n pills. He takes one aspirin at a time, randomly from one of the two boxes. After a while, he realizes that one box is empty.

- a) Find the probability of event A : when he notices that one box is empty, there are k ($k \leq n$) pills left in the other box.
- b) Use part a) to find a formula for $S_n = C_{2n}^n + 2 \cdot C_{2n-1}^n + \dots + 2^n \cdot C_n^n$.

Bonus Problems:

8. An urn contains 4 white balls and 5 black balls. Four balls are randomly removed without replacement. If the second and the third removed balls are black (while the first and forth removed balls are white), then 4 dice are rolled. Otherwise, 3 dice are rolled. Find the probability that exactly two rolled numbers are divisible by 3.

9. Three students take a test consisting of 10 questions. Their probabilities of answering a question correctly are 0.5, 0.8 and 0.6, respectively. Each question is worth one point and no fractions of a point are given. If each student answers every question, find the probability that two students will pass the exam (passing grades are 5, 6, ..., 10) and one will fail (event A).